



ADVANCED.

DP#1 _ Sequences

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Running time < 0.005s

1. Fibonacci numbers

Running time < 0.003s

Fibonacci sequence is defined as follows: $F_1 = F_2 = 1$, $F_n = F_{n-1} + F_{n-2}$, $n > 2$

Input data format

In one line of input data recorded positive integer n ($1 \leq n \leq 45$).

Output data format

Print a single number F_n .

Example

Input

3

Output

2

2. Ball on a ladder

Running time < 0.016s

At the top of the ladder, containing N steps, there is a ball that starts to jump on them down to the ground. The ball can jump to the next step, by step through one or 2. (That is, if the ball is on the eighth step, he can move to the fifth, sixth or seventh.) Determine the number of possible "routes" ball from the top to the ground.

Input data format

Introduced one number $0 < N < 31$.

Output data format

Print a single number - the number of routes.

Example

Input

4

Output

7



3. The last digit of the Fibonacci numbers

Fibonacci sequence is defined as follows: $F_0 = F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$. Write a program to calculate the last digit of n^{th} term of the sequence.

Input data format

In one line of input data recorded positive integer n ($1 \leq n \leq 1000$).

Output data format

Print the last digit of the number of F_n .

Example

Input

6

Output

3

4. Simple sequence

Calculate the n -th term of the sequence is given by:

$$a_{2n} = a_n + a_{n-1},$$

$$a_{2n+1} = a_n - a_{n-1},$$

$$a_0 = a_1 = 1.$$

Input data format

Introduce one integer n ($1 \leq n \leq 1000$).

Output data format

Print a single number a_n .

Example

Input

7

Output

-2

5. Cunning sequence

Calculate the n^{th} term of the sequence is given by:

$$a_{2n} = a_{n+1} \quad (n > 0),$$

$$a_{2n+1} = a_{2n} - a_n \quad (n > 0),$$

$$a_0 = 1, a_1 = 1.$$

introduce a positive integer n , does not exceed 1000.